RobiOmni
User Manual

Four-Wheel Drive and Four-Wheel Steering
Omnidirectional Chassis



Version V3.0.0 | English Edition

Safety Notice

Safety notice: This section contains important safety information for RobiOmni products. Before powering on and testing the chassis for the first time, read these instructions carefully and familiarize yourself with its operation. Pay particular attention to warning symbols.

Revision History

Version	Revision date	Changes
V1.0	2025.5.09	First edition
V2.0	2025.10.20	Mechanical design improvements
V3.0.0	2026.09.01	Third edition; protocol updates

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1 RobiOmni Chassis Overview

1.1 Product Overview

The RobiOmni series is a versatile, multimode mobile chassis with four-wheel drive and four-wheel steering, designed for indoor and outdoor campus environments. Typical applications include surveying and mapping, factory transport, and agriculture. Eight motors provide propulsion and steering: four drive motors and four steering motors. An ARM-based vehicle control unit (VCU) precisely controls vehicle speed and steering angle and provides real-time linear velocity and steering feedback. CAN communication allows a host computer to command the vehicle and read its status. The platform supports the development of autonomous driving, intelligent cruise control, and obstacle avoidance. ROS-based CAN communication packages are also provided to accelerate application development.



1.2 Advantages and Features

1.2.1 Omnidirectional Mobility

The RobiOmni chassis can move forward, backward, sideways, and diagonally, as well as turn left and right, providing high flexibility and maneuverability.

1.2.2 Precise Control

Each wheel can steer independently. Controlling wheel speed and direction enables precise positioning and motion, allowing the robot to maneuver in confined spaces.

1.2.3 Standard Communication Protocol

In addition to real-time remote control, the RobiOmni chassis provides a standard CAN control protocol. This supports real-time chassis monitoring and commands for motion and motor speed, simplifying integration with host computers and autonomous driving systems.

1.2.4 Expandability

The VCU provides RS232 serial or CAN communication expansion. Additional control boards can be integrated to support different tasks and applications.

1.2.5 Efficient Steering

Four independently steerable wheels enable faster, smoother turning, a smaller turning radius, and improved maneuverability and motion efficiency.

1.2.6 Versatile Applications

The RobiOmni series supports customization, including appearance, patent licensing, payload, and speed requirements. Its flexible motion and precise control suit indoor robots, automated guided vehicles, logistics vehicles, and other applications.

1.2.7 Key Features

1. The RobiOmni four-wheel-drive, four-wheel-steering omnidirectional chassis combines flexibility, precision, efficiency, and stability for a wide range of robotics applications.

2. The chassis supports standard CAN communication and provides 24V, 19V, and 12V power outputs for sensor integration. The quick-release battery supports convenient replacement and charging.

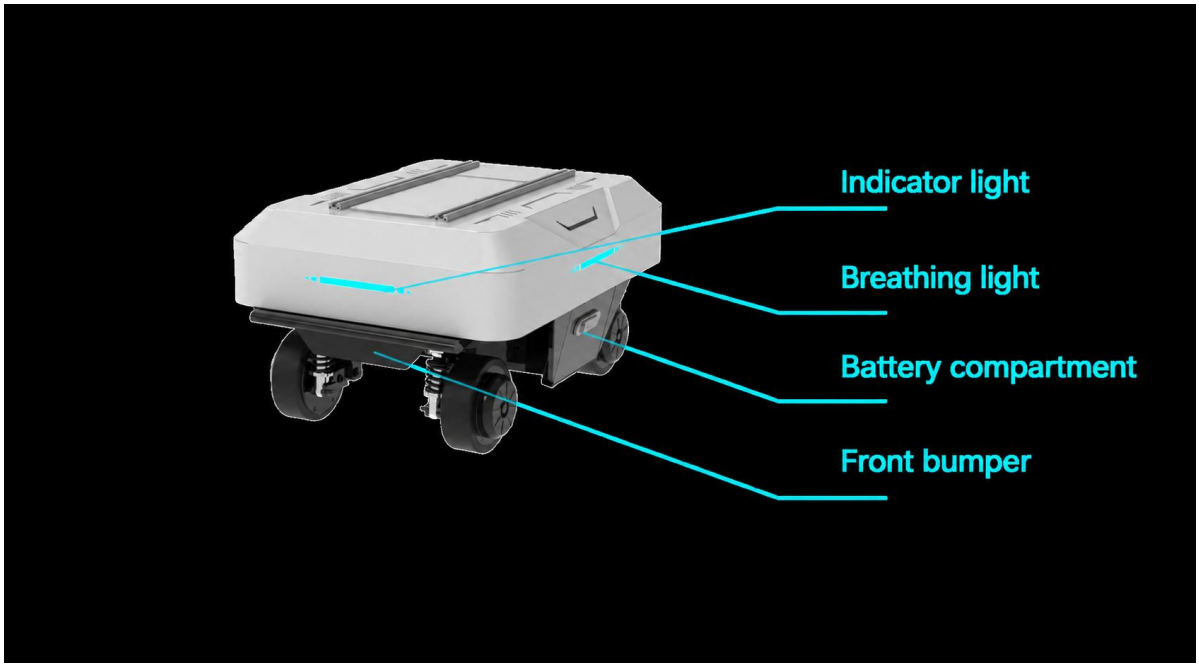
3. ROS support and standard interface packages help engineers and developers get started quickly.

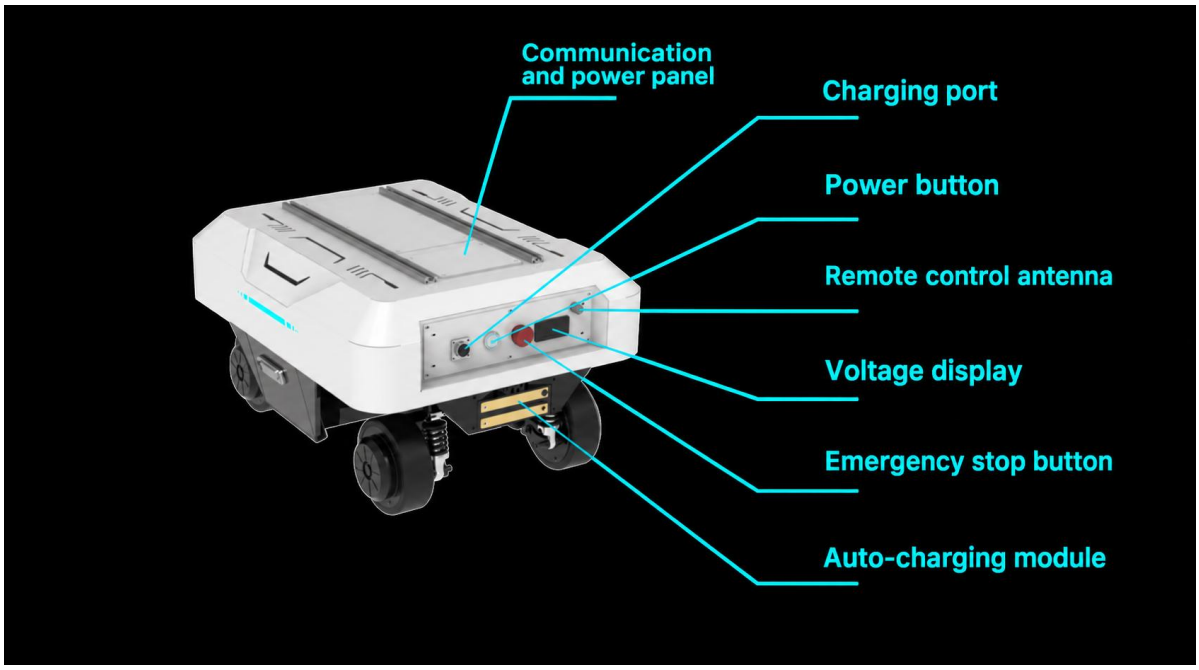
1.3 Accessories

Included Accessories			
RobiOmni chassis	1 set		
Lithium battery charger (220V AC)	1 set		
External power output module	1 set		
2.4GHz RC transmitter	1 set		
Power connectors	1 set		
Optional Accessories			
Laser navigation module	1 set	RGB-D camera	1 set
Multibeam LiDAR	1 set	RTK module	1 set
IMU	1 set	Navigation computer	1 set
Sensor mounting frame with display	1 set		

2 Chassis Components

2.1 Component Identification





Charging port: Connect the charger here. Charge promptly when battery voltage falls below 21V (approximately 20% SOC). Full-charge voltage is 25.2V. Stop using the chassis below 19V. Excessively low voltage triggers BMS undervoltage protection.

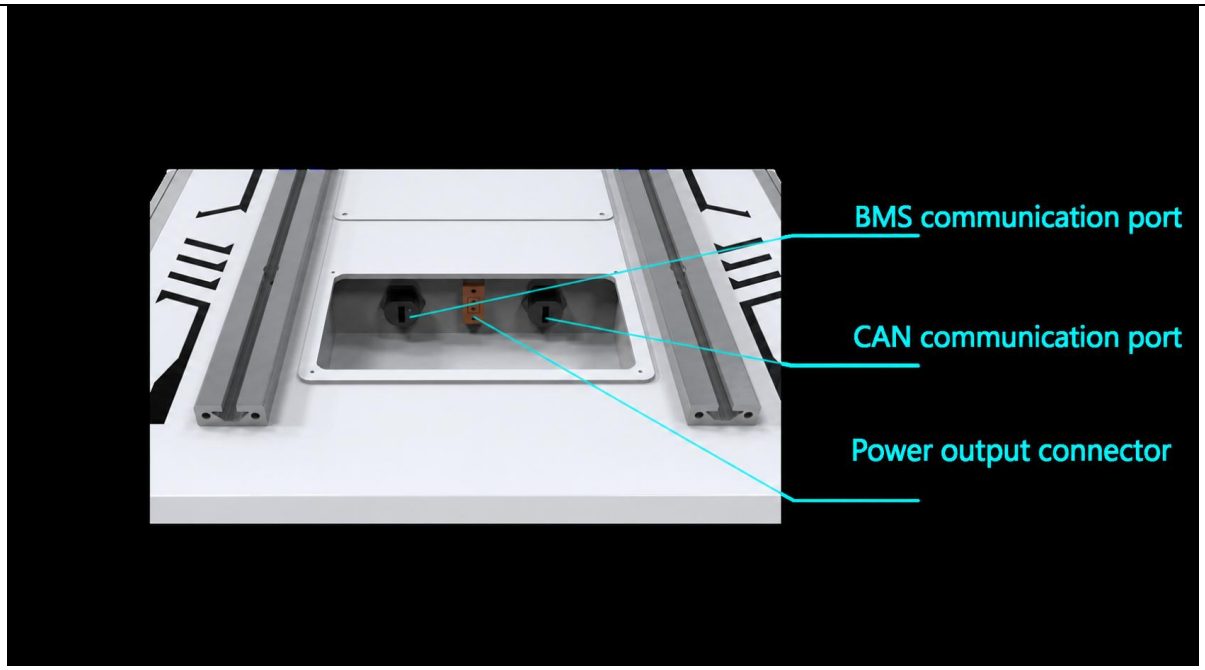
Power button: Press to enable power at the chassis output connectors; switch off to stop output. The output voltages are marked beside the connectors under the small chassis cover. Check polarity and never reverse the connections.

Emergency stop button: Press to stop the chassis in an emergency. Movement is possible only after this button is released. Turn it clockwise to release and reset the emergency stop.

Voltage display: Shows the current battery voltage when the chassis is powered on.

Remote control antenna: Receives control signals from the RC transmitter.

2.2 Connector Panel

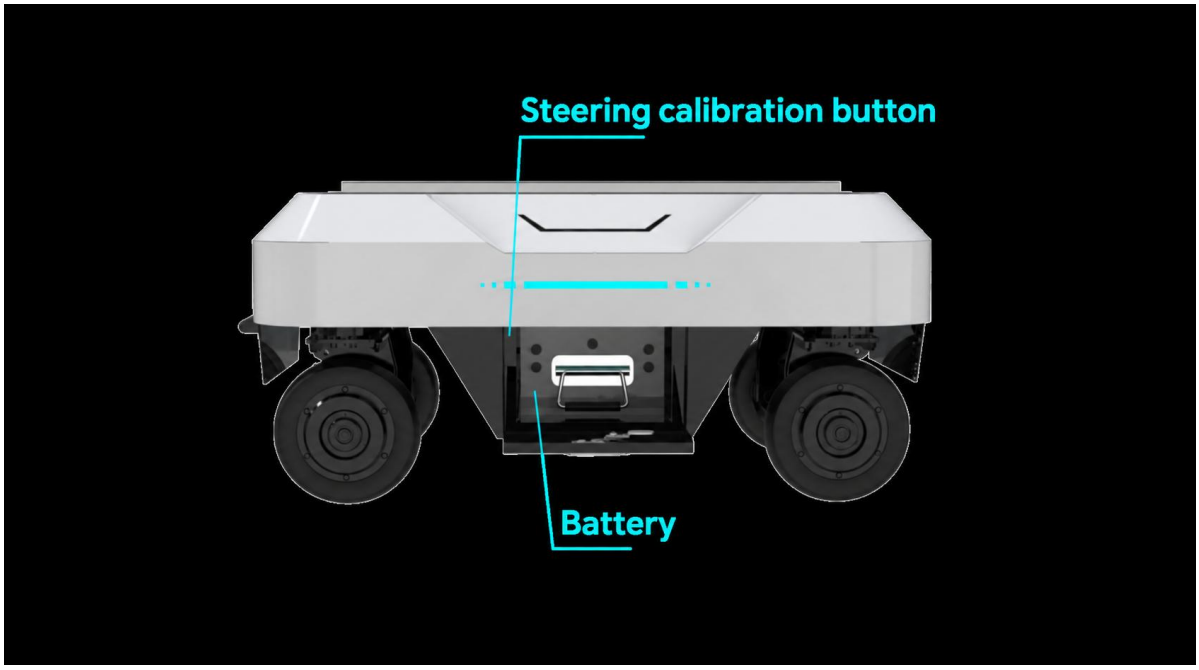


BMS communication port: USB-A receptacle

Power output connector: XT60 female

CAN communication port: USB-B receptacle

2.3 Battery Compartment



The chassis battery is removable. Press the battery compartment cover to open it. Battery connections are defined below.

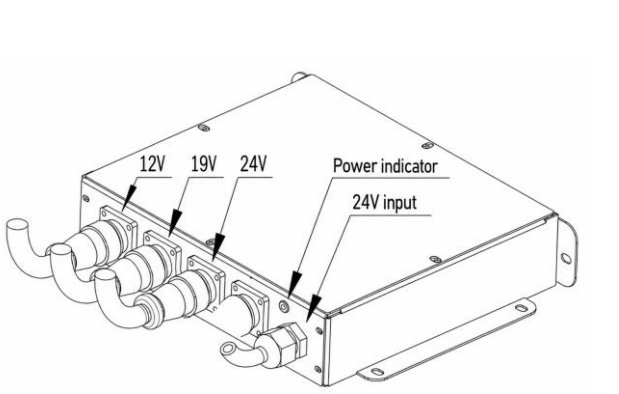
Battery discharge connector: XT90 female

Battery charging connector: XT60 female

BMS interface: USB. Reconnect all three plugs after replacing the battery; otherwise normal operation may be affected.

Calibration button: Used to calibrate steering. See the operating instructions later in this manual.

2.4 Power Distribution Box





24V Connector Pinout

Connector model	Pin	Function
WEIPU WS20-5P	1	Positive
Output rating	2	Positive

Connector model	Pin	Function
24V 10A	3	Negative
	4	Negative
	5	NC

19V Connector Pinout

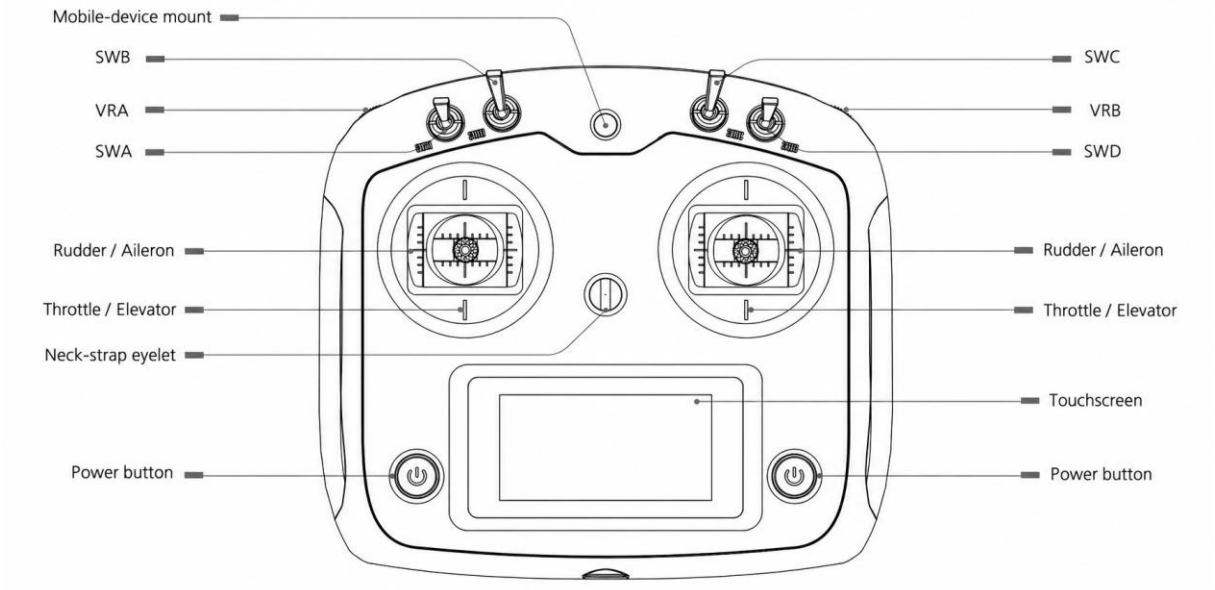
Connector model	Pin	Function
WEIPU WS20-4P	1	Positive
Output rating	2	Positive
19V 10A	3	Negative
	4	Negative

12V Connector Pinout

Connector model	Pin	Function
WEIPU WS20-3P	1	Positive
Output rating	2	Negative
12V 15A	3	NC

3 Operating Instructions

3.1 RC Transmitter Instructions



Power buttons: Press the left and right power buttons simultaneously to turn the transmitter on or off.

SWA: Speed selection. Upper position selects full speed; lower position selects half speed.

SWB: Control mode. Lower position selects remote control; middle position selects CAN/RS232 control; upper position initiates automatic charging.

SWC: Lower position selects FTFD mode; middle position selects Ackermann mode; upper position selects double Ackermann mode.

SWD: Remote emergency stop. Upper position activates the emergency stop; lower position releases it. While the transmitter is on and SWD is in the upper position, neither remote control nor CAN can move the chassis.

Operation with SWC in the Lower Position

Left stick: Move forward or backward to drive forward or reverse. Move diagonally to drive in the corresponding diagonal direction, or left/right to move sideways.

Right stick: Move left or right to turn. Hold the stick in one direction to rotate in place: left for counterclockwise rotation and right for clockwise rotation.

Operation with SWC in the Middle Position

Left stick: Move forward/backward to drive forward/reverse; move left/right to rotate in place.

Right stick: Move left/right to steer left/right.

VRA and VRB are not used at present.

The buttons on the left and right sides of the transmitter rear have no assigned function.

3.2 Chassis Operation and Functions

3.2.1 Power On

1. Press the power button on the switch panel. One buzzer tone indicates that startup is complete. A continuous buzzer indicates a fault; stop using the chassis.

2. Check the voltage display. If voltage is below 19V, stop using the chassis and charge it as described below.

3.2.2 Remote Control

Step 1. Power on the chassis and release its emergency stop.

Step 2. Turn on the RC transmitter and wait for the touchscreen to illuminate. Before startup, set SWA, SWB, SWC, and SWD to their upper positions. If they are not all up, the touchscreen displays a warning; move them all up to continue.

Step 3. Set SWB down for remote control. Set SWC down for omnidirectional steering or to the middle for Ackermann steering. Set SWD down to release the remote emergency stop.

Step 4. Use the left stick for forward/reverse motion and the right stick for left/right steering. Use SWA to select high or low speed.

3.2.3 Emergency Stop

Press the red emergency stop button at the rear of the chassis, or move SWD on the transmitter to its upper position.

3.2.4 Power Off

Press the power button on the switch panel. Except in an emergency, do not switch off power while the chassis is moving.

3.2.5 Charging



The chassis uses a nominal 24V battery with an operating voltage range of 18-25.2V. Stop using it and recharge when voltage falls below 19V. Follow these steps:

1. Switch off the chassis.
2. Connect the charger to the chassis charging port first, then connect the charger to a 220V outlet.
3. The charger indicator is red during charging and turns green when charging is complete.
4. To interrupt charging, disconnect the 220V supply first, then unplug the charger from the chassis.

3.2.6 Automatic Charging

If the chassis is fitted with an automatic charging module, it can dock for charging while the battery still has enough power for normal movement (a voltage above 24V is recommended). Follow the procedure below.

Host-Controlled Charging

1. Start the chassis using the normal procedure.
2. Use remote control to stop the chassis approximately 1m directly in front of the charging dock, then turn off the transmitter.
3. Send the start-charging command from the host computer.
4. The chassis automatically approaches the dock and begins charging.
5. To exit automatic charging, send the stop-charging command from the host computer.

Remote-Controlled Charging

1. Start the chassis using the normal procedure.
2. Use remote control to stop the chassis approximately 1m directly in front of the charging dock.
3. Move SWB to its upper position to select automatic charging.
4. The chassis automatically approaches the dock and charges. The transmitter may now be turned off.

5. To exit automatic charging, move SWB back down to remote-control mode.

3.2.7 Steering Calibration

If steering is abnormal after startup or the wheels cannot align straight ahead, use the following calibration procedure:

1. Power on the chassis normally, then press its red emergency stop button.

2. Manually align the wheels straight ahead, or use CAN commands to align the front wheels.

3. Press and hold the calibration button (see the battery compartment section) until the chassis beeps, then release it.

4. Release the emergency stop and restart chassis power to resume normal control.

Note: Steering is calibrated before shipment and normally does not require recalibration.

4 Chassis Communication Protocol

4.1 Communication Format (RobiOmni CAN Protocol V2.2)

Baud rate	Frame format	Frame type	Data format
500Kbs	Standard frame	Data frame	Little-endian

4.1.1 Host Command Frame Format

CAN_ID	Length	Byte 0	Byte 1-7
0x01	8	Command	Data

4.2 Host Commands

4.2.1 Set Motion Mode and Speed

Byte	Description		Data type	Notes	
Byte 0	0x01		Unsigned int8	Command	
Byte 1	Motion mode		Unsigned int8	0x00: Ackermann / rotation in place 0x01: FTFD mode 0x02: Double Ackermann mode	
	Ackermann mode	FTFD mode		Ackermann mode	FTFD mode
Byte 2	Linear speed, low byte	Vx, low byte	int16	Unit: mm/s	
Byte 3	Linear speed, high byte	Vx, high byte			
Byte 4	Steering angle, low byte	Vy, low byte	int16	Unit: 0.01 degrees (Left positive; right negative)	Unit: mm/s (Left positive; right negative)
Byte 5	Steering angle, high byte	Vy, high byte			

Byte	Description		Data type	Notes
Byte 6	Angular velocity, low byte	Vz, low byte	int16	Unit: 0.001rad/s (counterclockwise positive)
Byte 7	Angular velocity, high byte	Vz, high byte		

4.2.2 Enable or Disable Automatic Charging

Byte	Description	Data type	Notes
Byte 0	0x02	Unsigned int8	Data
Byte 1	0x00/0x01/0x02	Unsigned int8	0x00: Disable charging 0x01: Enable infrared charging 0x02: Enable laser charging
Byte 2 ~ Byte 7	Reserved		Reserved

4.2.3 Clear Collision Status

Byte	Description	Data type	Notes
Byte 0	0x20	Unsigned int8	Command
Byte 1	0x01	Unsigned int8	1: Clear
Byte 2 ~ Byte 7	Reserved		Reserved

4.2.4 Turn Lighting On or Off

Byte	Description	Data type	Notes
Byte 0	0x40	Unsigned int8	Command
Byte 1	0x00/0x01	Unsigned int8	0x00: Off 0x01: On
Byte 2 ~ Byte 7	Reserved		Reserved

4.3 Feedback Data

4.3.1 Motion Feedback (ID: 0x10), Every 20ms

Byte	Description	Data type	Notes
Byte 0	0x01	Unsigned int8	Command

Byte	Description		Data type	Notes	
Byte 1	Motion mode		Unsigned int8	0x00: Ackermann / rotation in place 0x01: FTFD mode 0x02: Double Ackermann mode	
	Ackermann mode	FTFD mode		Ackermann mode	FTFD mode
Byte 2	Linear speed, low byte	Vx, low byte	int16	Unit: mm/s	
Byte 3	Linear speed, high byte	Vx, high byte			
Byte 4	Steering angle, low byte	Vy, low byte	int16	Unit: 0.01 degrees (left positive; right negative)	Unit: mm/s (Left positive; right negative)
Byte 5	Steering angle, high byte	Vy, high byte			
Byte 6	Angular velocity, low byte	Vz, low byte	int16	Unit: 0.001rad/s (counterclockwise positive)	
Byte 7	Angular velocity, high byte	Vz, high byte			

4.3.2 System Status (ID: 0x11), Every 20ms

Byte	Description	Data type	Notes
Byte 0 Byte 1	Battery voltage	Unsigned int16	Unit: 0.1V
Byte 2	Control mode	Unsigned int8	0x00: Idle 0x01: Remote control 0x02: CAN control 0x03: External control
Byte 3	Status: Bit0: Remote control online Bit1: Hardware emergency stop Bit2: Remote emergency stop Bit3: Software emergency stop Bit4-Bit7: Reserved	Unsigned int8	1: Online / active 0: Offline / inactive

Byte	Description	Data type	Notes
Byte 4	Drive faults Bit0: Front-left drive motor Bit1: Front-right drive motor Bit2: Rear-left drive motor Bit3: Rear-right drive motor Bit4: Front-left steering motor Bit5: Front-right steering motor Bit6: Rear-left steering motor Bit7: Rear-right steering motor	Unsigned int8	0: Normal 1: Fault
Byte 5	Encoder faults Bit0: Front-left drive motor Bit1: Front-right drive motor Bit2: Rear-left drive motor Bit3: Rear-right drive motor Bit4: Front-left steering motor Bit5: Front-right steering motor Bit6: Rear-left steering motor Bit7: Rear-right steering motor	Unsigned int8	0: Normal 1: Fault
Byte 6	Absolute encoder errors: Bit0: Front-left absolute encoder error Bit1: Front-right absolute encoder error Bit2: Rear-left absolute encoder error Bit3: Rear-right absolute encoder error Bit4-Bit7: Reserved	Unsigned int8	0: Normal 1: Error
Byte 7	Battery percentage	Unsigned int8	Calculated from voltage; range 0 ~ 100%

4.3.3 Drive Motor Status (ID: 0x12), Every 20ms

Byte	Description	Data type	Notes
Byte 0	0x01	Unsigned int8	Fixed
Byte 1	Motor ID	Unsigned int8	0x01: Front-left drive motor 0x02: Front-right drive motor 0x03: Rear-left drive motor 0x04: Rear-right drive motor
Byte 2	Motor speed, low byte	int16	Unit: mm/s
Byte 3	Motor speed, high byte		

Byte	Description	Data type	Notes
Byte 4	Encoder count, byte 0 (LSB)	Unsigned int32	
Byte 5	Encoder count, byte 1		
Byte 6	Encoder count, byte 2		
Byte 7	Encoder count, byte 3 (MSB)		

Example: 01 01 EC 00 DC F4 07 00 (front-left motor; speed: 236mm/s; encoder count: 521436)

4.3.4 Steering Motor Status (ID: 0x13), Every 20ms

Byte	Description	Data type	Notes
Byte 0	0x01	Unsigned int8	Fixed
Byte 1	Motor ID	Unsigned int8	0x01: Front-left drive motor 0x02: Front-right drive motor 0x03: Rear-left drive motor 0x04: Rear-right drive motor
Byte 2	Motor angle, low byte	int16	Unit: 0.01 degrees
Byte 3	Motor angle, high byte		
Byte 4 - Byte 7	Reserved		Reserved

Example as above: 01 = front-left motor; 02 = front-right motor; 03 = rear-left motor; 04 = rear-right motor.

4.3.5 Automatic Charging Status (ID: 0x14), Every 100ms

Byte	Description	Data type	Notes
Byte 0	Charging module offline	Unsigned int8	0x00: Online 0x01: Offline
Byte 1	Current charging mode	Unsigned int8	0x00: Idle 0x01: Infrared mode 0x02: Laser mode
Byte 2	Charging module status: Bit0: Dock limit switch status Bit1: Charging contact relay status Bit2: Charging contact voltage detection	Unsigned int8	0: Released / no voltage detected 1: Pressed / voltage detected

Byte	Description	Data type	Notes
Byte 3	Infrared charging status	Unsigned int8	0x00: Searching for center 0x01: Center found 0x02: Signal lost 0x03: Copper contacts connected 0x04: Docking successful 0x05: Docking error 0x07: Exiting charging 0x08: Charging task complete / disabled
Byte 4 - Byte 7	Reserved		Reserved

4.3.6 RC Joystick Feedback (ID: 0x15), Every 50ms

Byte	Description	Data type	Notes
Byte 0 Byte 1	Right stick, left/right	Int16	Left: -584 Center: 0 Right: 583
Byte 2 Byte 3	Right stick, up/down	Int16	Down: -584 Center: 0 Up: 583
Byte 4 Byte 5	Left stick, up/down	Int16	Down: -584 Center: 0 Up: 583
Byte6 Byte 7	Left stick, left/right	Int16	Left: -584 Center: 0 Right: 583

4.3.7 RC Dial and Switch Feedback (ID: 0x16), Every 50ms

Byte	Description	Data type	Notes
Byte 0 Byte 1	Left dial, forward/backward	Int16	Left: -584 Center: 0 Right: 583
Byte 2 Byte 3	Right dial, forward/backward	Int16	Down: -584 Center: 0 Up: 583

Byte	Description	Data type	Notes
Byte 4	Switches	Unsigned int8	Bit assignment Bit 0 - bit 1 : SWA Bit 2 - bit 3 : SWB Bit 4 - bit 5 : SWC Bit 6 - bit 7 : SWD Values 0: Lower 1: Middle 2: Upper 3: Reserved
Byte 5	Remote control status	Unsigned int8	Bit 0: Offline Bit 1-Bit 7: Reserved 0: No 1: Yes
Byte6 Byte 7	Reserved	Unsigned int8	Reserved

4.3.8 System Status 2 (ID: 0x30), Every 100ms

Byte	Description	Data type	Notes
Byte 0 Byte 1	Status 2 Bit0: Front bumper triggered Bit1: Rear bumper triggered Bit2: Lighting status Bit3-Bit15: Reserved	Unsigned int16	Bit = 0: Inactive / off Bit = 1: Active / on
Byte 2	Drive offline status: Bit0: Front-left drive motor offline Bit1: Front-right drive motor offline Bit2: Rear-left drive motor offline Bit3: Rear-right drive motor offline Bit4: Front-left steering motor offline Bit5: Front-right steering motor offline Bit6: Rear-left steering motor offline Bit7: Rear-right steering motor offline	Unsigned int8	Bit = 0: Normal Bit = 1: Offline

Byte	Description	Data type	Notes
Byte 3~ Byte 7	Reserved		Reserved

4.3.9 Steering Motor Current (ID: 0x40), Every 20ms

Byte	Description	Data type	Notes
Byte 0	Front-left steering motor current, low byte	int16	Not specified
Byte 1	Front-left steering motor current, high byte		
Byte 2	Front-right steering motor current, low byte	int16	
Byte 3	Front-right steering motor current, high byte		
Byte 4	Rear-left steering motor current, low byte	int16	
Byte 5	Rear-left steering motor current, high byte		
Byte 6	Rear-right steering motor current, low byte	int16	
Byte 7	Rear-right steering motor current, high byte		

4.3.10 Drive Motor Current (ID: 0x41), Every 20ms

Byte	Description	Data type	Notes
Byte 0	Front-left drive motor current, low byte	int16	Not specified
Byte 1	Front-left drive motor current, high byte		
Byte 2	Front-right drive motor current, low byte	int16	
Byte 3	Front-right drive motor current, high byte		
Byte 4	Rear-left drive motor current, low byte	int16	
Byte 5	Rear-left drive motor current, high byte		
Byte 6	Rear-right drive motor current, low byte	int16	
Byte 7	Rear-right drive motor current, high byte		

4.3.11 Drive and Motor Faults 1 (ID: 0x42), Every 100ms

Byte	Description	Data type	Notes
Byte 0	Front-left drive motor current, low byte	Unsigned int16	Bit[0]: Abnormal supply voltage Bit[1]: Reserved Bit[2]: Reserved Bit[3]: Reserved Bit[4]: Reserved Bit[5]: FOC process timeout Bit[6]: Drive overtemperature Bit[7]: Reserved Bit[8]-Bit[15]: Reserved
Byte 1	Front-left drive motor current, high byte		
Byte 2	Front-right drive motor current, low byte	Unsigned int16	
Byte 3	Front-right drive motor current, high byte		
Byte 4	Rear-left drive motor current, low byte	Unsigned int16	
Byte 5	Rear-left drive motor current, high byte		
Byte 6	Rear-right drive motor current, low byte	Unsigned int16	
Byte 7	Rear-right drive motor current, high byte		

4.3.12 Drive and Motor Faults 2 (ID: 0x43), Every 100ms

Byte	Description	Data type	Notes
Byte 0	Front-left drive motor fault, low byte	Unsigned int16	Bit[0]: Internal error Bit[1]: Encoder ABZ signal error Bit[2]: Encoder UVW signal error Bit[3]: Encoder count error Bit[4]: Drive overtemperature Bit[5]: Drive bus overvoltage Bit[6]: Drive bus undervoltage Bit[7]: Drive output short circuit Bit[8]: Braking resistor overtemperature Bit[9]: Position following error exceeds limit Bit[10]: Reserved Bit[11]: I2*T fault (drive or motor overload) Bit[12]: Speed following error exceeds limit; Bit[13]: Motor overtemperature Bit[14]: Motor search error (communication encoder) Bit[15]: Communication offline alarm
Byte 1	Front-left drive motor fault, high byte		
Byte 2	Front-right drive motor fault, low byte	Unsigned int16	
Byte 3	Front-right drive motor fault, high byte		
Byte 4	Rear-left drive motor fault, low byte	Unsigned int16	
Byte 5	Rear-left drive motor fault, high byte		
Byte 6	Rear-right drive motor fault, low byte	Unsigned int16	
Byte 7	Rear-right drive motor fault, high byte		

4.3.13 Battery Status (ID: 0x50), Every 100ms

Byte	Description	Data type	Notes
Byte 0	Status flags	Unsigned int8	bit0: Offline bit1: Alarm bit2: Charging MOS on bit3: Discharging MOS on bit4-bit7: Reserved bit = 0: No; bit = 1: Yes
Byte 1	Charge/discharge status	Unsigned int8	0: Idle 1: Charging 2: Discharging
Byte 2	Battery voltage, low byte	Unsigned int16	Unit: 0.1V
Byte 3	Battery voltage, high byte		

Byte	Description	Data type	Notes
Byte 4	Battery current, low byte	Unsigned int16	Unit: 0.1A
Byte 5	Battery current, high byte		
Byte 6	Battery SOC, low byte	Unsigned int16	Unit: 0.1%
Byte 7	Battery SOC, high byte		

4.3.14 Battery Information (ID: 0x51), Every 1000ms

Byte	Description	Data type	Notes
Byte 0	Remaining capacity, low byte	Unsigned int16	Unit: 0.01Ah
Byte 1	Remaining capacity, high byte		
Byte 2	Maximum cell temperature	Unsigned int8	Unit: degrees Celsius
Byte 3	Minimum cell temperature	Unsigned int8	
Byte 4~Byte 7	Reserved		Reserved

4.3.15 Battery Information (ID: 0x52), Every 1000ms

Byte	Description	Data type	Notes
Byte 0	Alarm information	Unsigned int8	bit0: Cell overvoltage, level 1 bit1: Cell overvoltage, level 2 bit2: Cell undervoltage, level 1 bit3: Cell undervoltage, level 2; bit4: Pack overvoltage, level 1 bit5: Pack overvoltage, level 2 bit6: Pack undervoltage, level 1 bit7: Pack undervoltage, level 2 bit = 0: No; bit = 1: Yes

Byte	Description	Data type	Notes
Byte 1	Alarm information	Unsigned int8	bit0: Charging overtemperature, level 1 bit1: Charging overtemperature, level 1 bit2: Charging undertemperature, level 1 bit3: Charging undertemperature, level 2 bit4: Discharging overtemperature, level 1 bit5: Discharging overtemperature, level 2 bit6: Discharging undertemperature, level 1 bit7: Discharging undertemperature, level 2 bit = 0: No; bit = 1: Yes
Byte 2	Alarm information	Unsigned int8	bit0: Charging overcurrent, level 1 bit1: Charging overcurrent, level 2 bit2: Discharging overcurrent, level 1 bit3: Discharging overcurrent, level 2 bit4: SOC too high, level 1 bit5: SOC too high, level 2 bit6: SOC too low, level 1 bit7: SOC too low, level 2 bit = 0: No; bit = 1: Yes
Byte 3	Alarm information	Unsigned int8	bit0: Cell voltage difference too large, level 1 bit1: Cell voltage difference too large, level 2 bit2: Cell temperature difference too large, level 1 bit3: Cell temperature difference too large, level 2 bit4-bit7: Reserved bit = 0: No; bit = 1: Yes

Byte	Description	Data type	Notes
Byte 4	Alarm information	Unsigned Int8	bit0: Charging MOS overtemperature warning bit1: Discharging MOS overtemperature warning bit2: Charging MOS temperature sensor fault bit3: Discharging MOS temperature sensor fault bit4: Charging MOS stuck on bit5: Discharging MOS stuck on bit6: Charging MOS open circuit bit7: Discharging MOS open circuit bit = 0: No; bit = 1: Yes
Byte 5	Alarm information	Unsigned int8	bit0: AFE acquisition IC fault bit1: Cell measurement offline bit2: Cell temperature sensor fault bit3: EEPROM storage fault bit4: RTC clock fault bit5: Precharge failed bit6: Vehicle communication fault bit7: Internal network communication module fault bit = 0: No; bit = 1: Yes
Byte 6	Alarm information	Unsigned int8	bit0: Current measurement module fault bit1: Internal pack voltage measurement fault bit2: Short-circuit protection fault bit3: Charging inhibited due to low voltage bit4-bit7: Reserved bit = 0: No; bit = 1: Yes
Byte 7	Alarm information	Unsigned int8	Fault code value

5 Precautions

5.1 Operating Environment

1. Operating temperature: -10 to 60 degrees Celsius, indoors or outdoors. Do not operate outside this range.
2. Operating relative humidity: 30%-90%.

3. Do not operate the chassis where corrosive or flammable gases are present.

4. Recommended operating altitude: no more than 2000m.

5.2 Mechanical Loading

1. Maximum vertical payload: 50kg.

2. Ensure the application meets the operating environment requirements.

3. Use the chassis on level ground or suitable standard indoor/outdoor surfaces.

5.3 External Electrical Connections

1. Measure output voltage and verify polarity before connecting external equipment. Never reverse polarity.

2. Do not exceed the rated output power of the power expansion board.

5.4 Other Precautions

1. Read the relevant manuals before using the chassis.

2. Switch off the chassis promptly if you notice a burning smell or any other abnormal condition.

3. During storage, charge the battery at least once a month to prevent deep discharge and battery damage.

6 Frequently Asked Questions

Q: Why does the RobiOmni chassis not move under remote control after startup?

A: Check that the transmitter is on and connected to the chassis, and that the chassis emergency stop is released. If these checks pass, follow Section 3.2 and confirm that the transmitter mode switches are set correctly.

Q: Why can the host computer not control the RobiOmni chassis?

A: Check whether the RC transmitter is on. Remote control normally has the highest priority. Turn the transmitter off, or move its control mode switch to the middle position. The middle position allows host communication while retaining the remote braking function.

Q: What should I do when the voltage display approaches the red range?

A: Recharge the chassis as soon as possible when the voltage approaches the red range.